

Sentiment Analysis of Tweets on Tech Products

SENTIMENT ANALYSIS



**Discovering people opinions, emotions and feelings about
a product or service**

Business Understanding

1

In the digital age, customer opinions on social media directly shape brand perception.

2

Apple and Google dominate tech markets; their online reputation influences purchase decisions

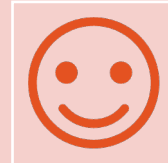
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Understanding sentiment helps align marketing, product design, and customer experience

Problem Statement



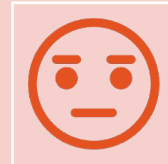
How do consumers feel about Apple vs Google products?



What themes drive positive and negative sentiment?



Can we predict public sentiment accurately using machine learning?



Which aspects influence consumer emotion the most (price, design, reliability, etc.)?

Objectives



Collect and clean Apple & Google tweet data



Build a predictive NLP model to classify sentiment



Identify top drivers of positive and negative sentiment



Deploy a Streamlit dashboard for live analysis and visualization

Data Understanding

Dataset: “Brands and Product Emotions” (CrowdFlower via Data.World)

We have approximately 9,000 tweets mentioning tech products

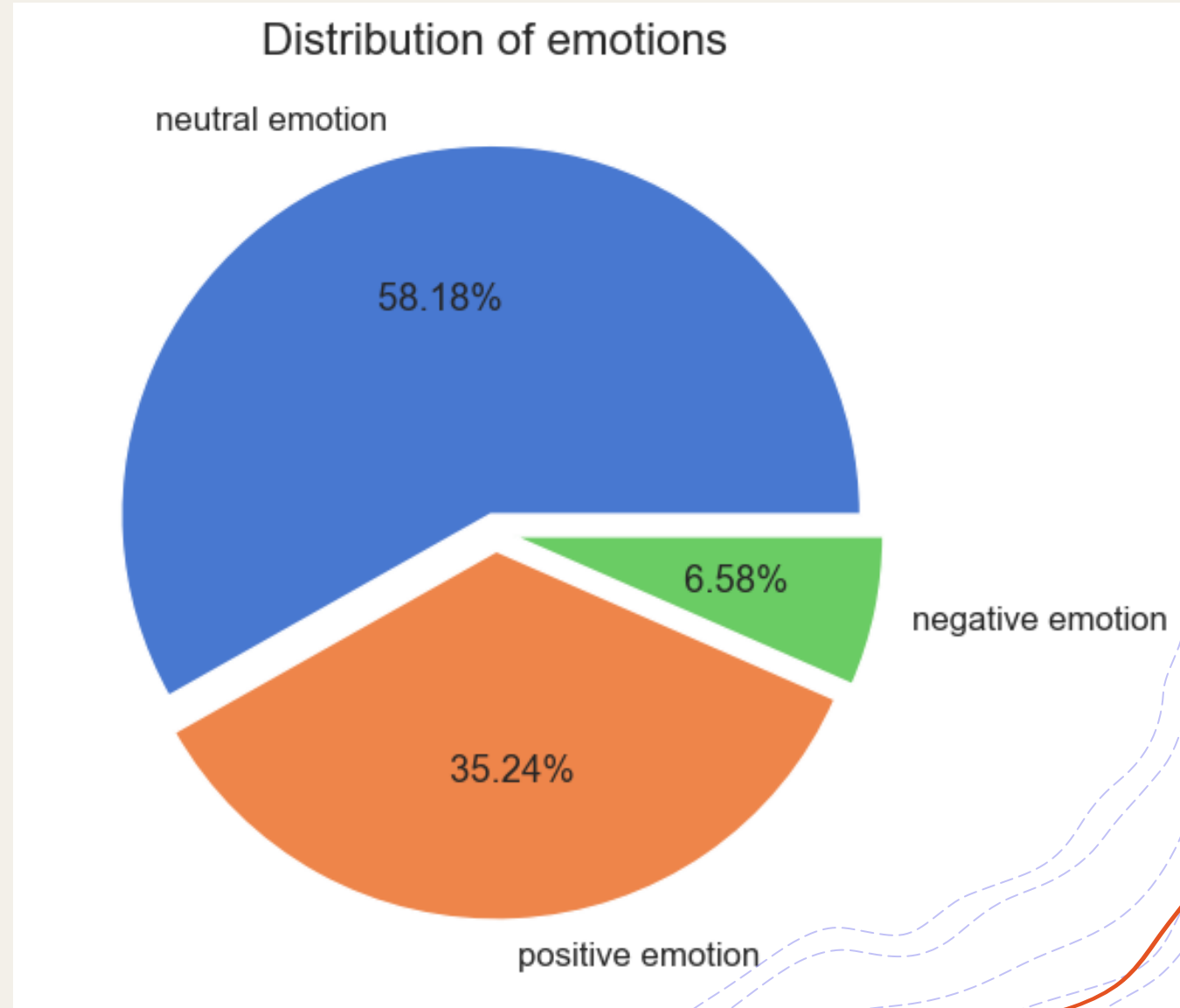
Variables:

1. tweet_text – user message
 2. emotion_in_tweet_is_directed_at – product/brand target
 3. is_there_an_emotion_directed_at_a_brand_or_product – sentiment label
-

Focused only on Apple & Google entries

Descriptive Analytics

- + Positive: 35%
- + Neutral: 58%
- + Negative: 7%
- + Apple tweets more emotionally expressive (strong positives & negatives)
- + Google tweets mostly neutral or product-oriented





Diagnostic Insights

- + Common positive terms: “love”, “smooth”, “design”, “ecosystem”
- + Common negative terms: “battery”, “price”, “bug”, “crash”
- + Apple sentiment linked to experience & design
- + Google sentiment linked to performance & reliability

Positive

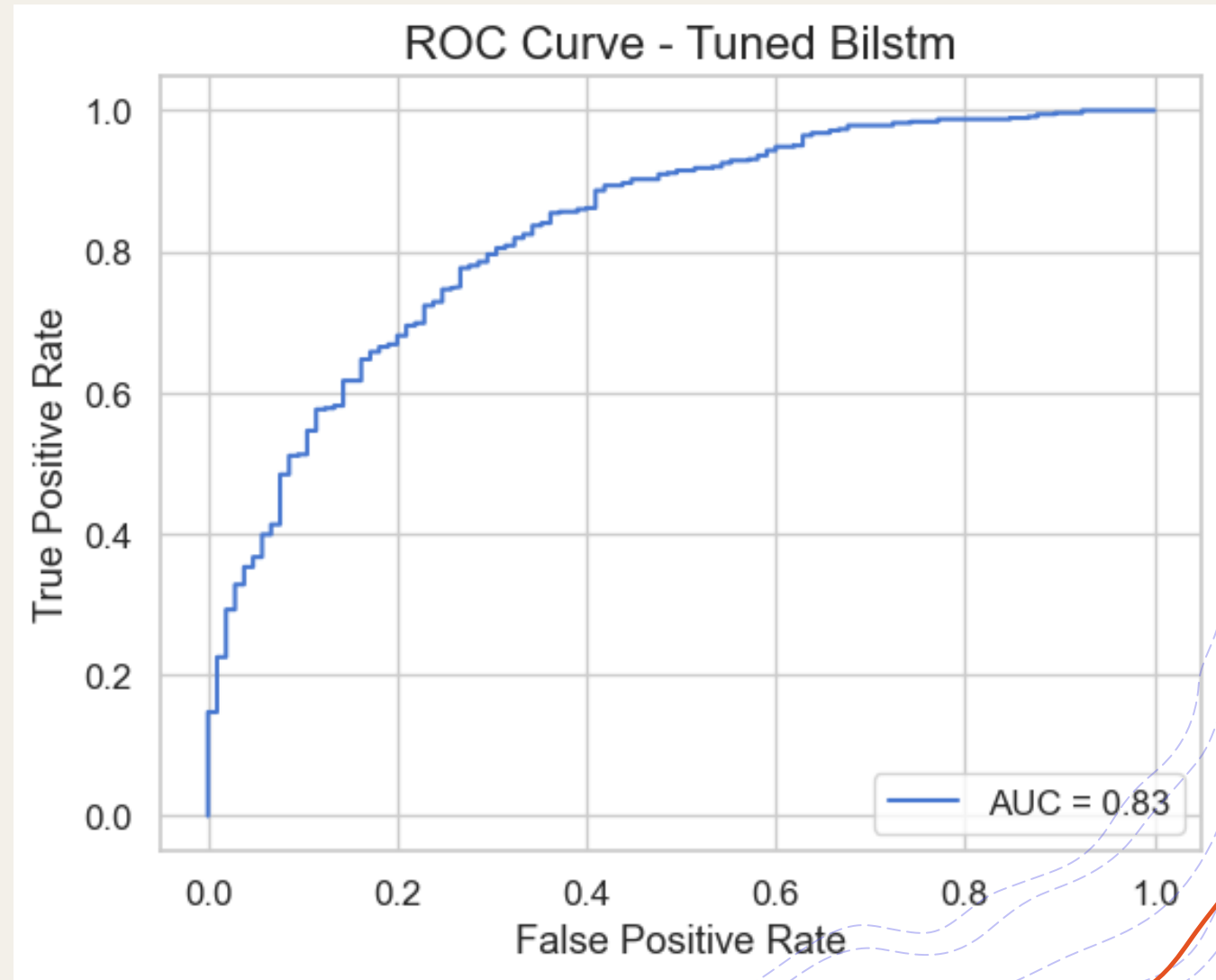


Negative



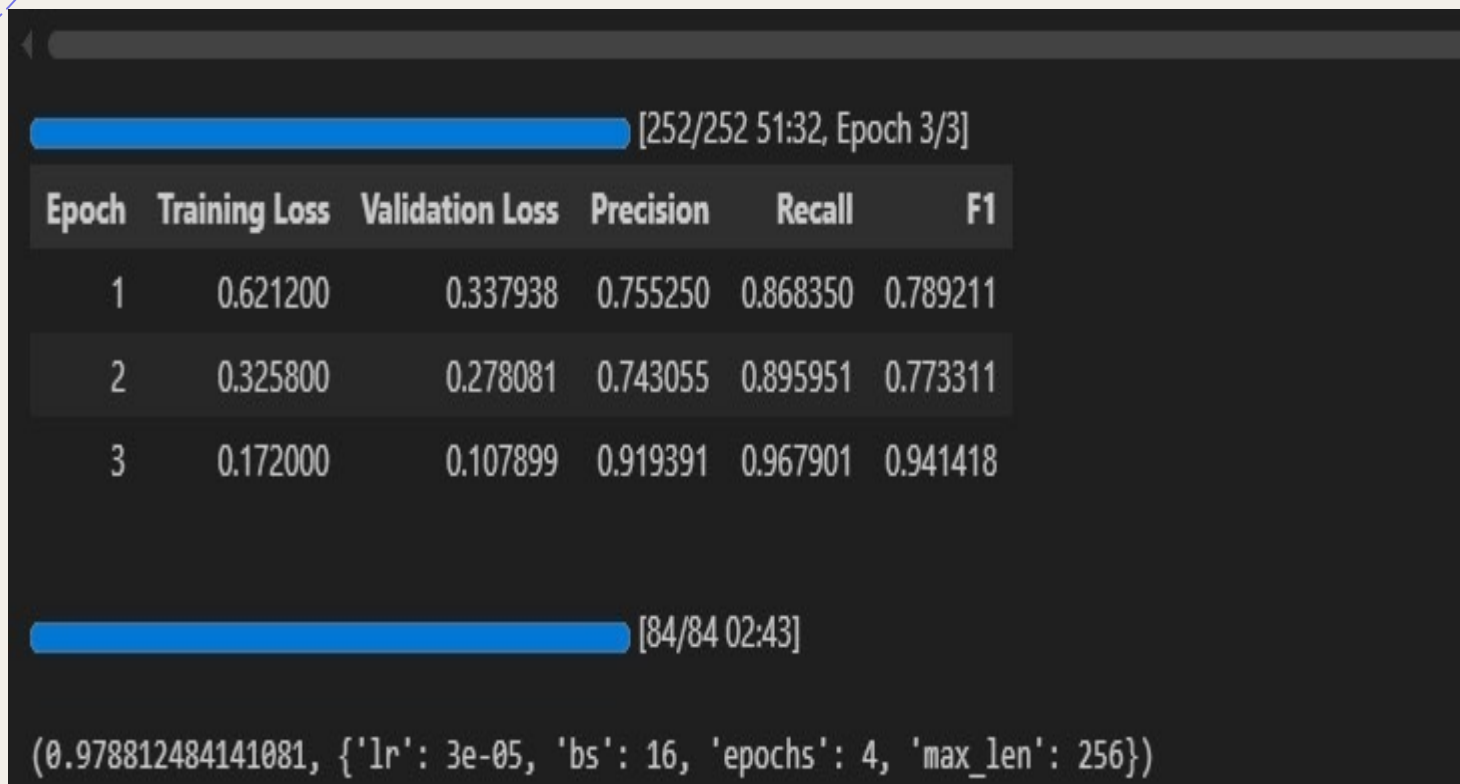
Predictive Modeling

- + Models tested:
 1. Logistic Regression (TF-IDF)
 2. Naive Bayes
 3. DistilBERT Transformer (fine-tuned)
- + Evaluation metrics: Accuracy, F1-Score, Precision, Recall
- + Addressed class imbalance via stratified sampling



Model Results

- + DistilBERT achieved high precision across all classes
- + Excellent handling of linguistic nuance and context



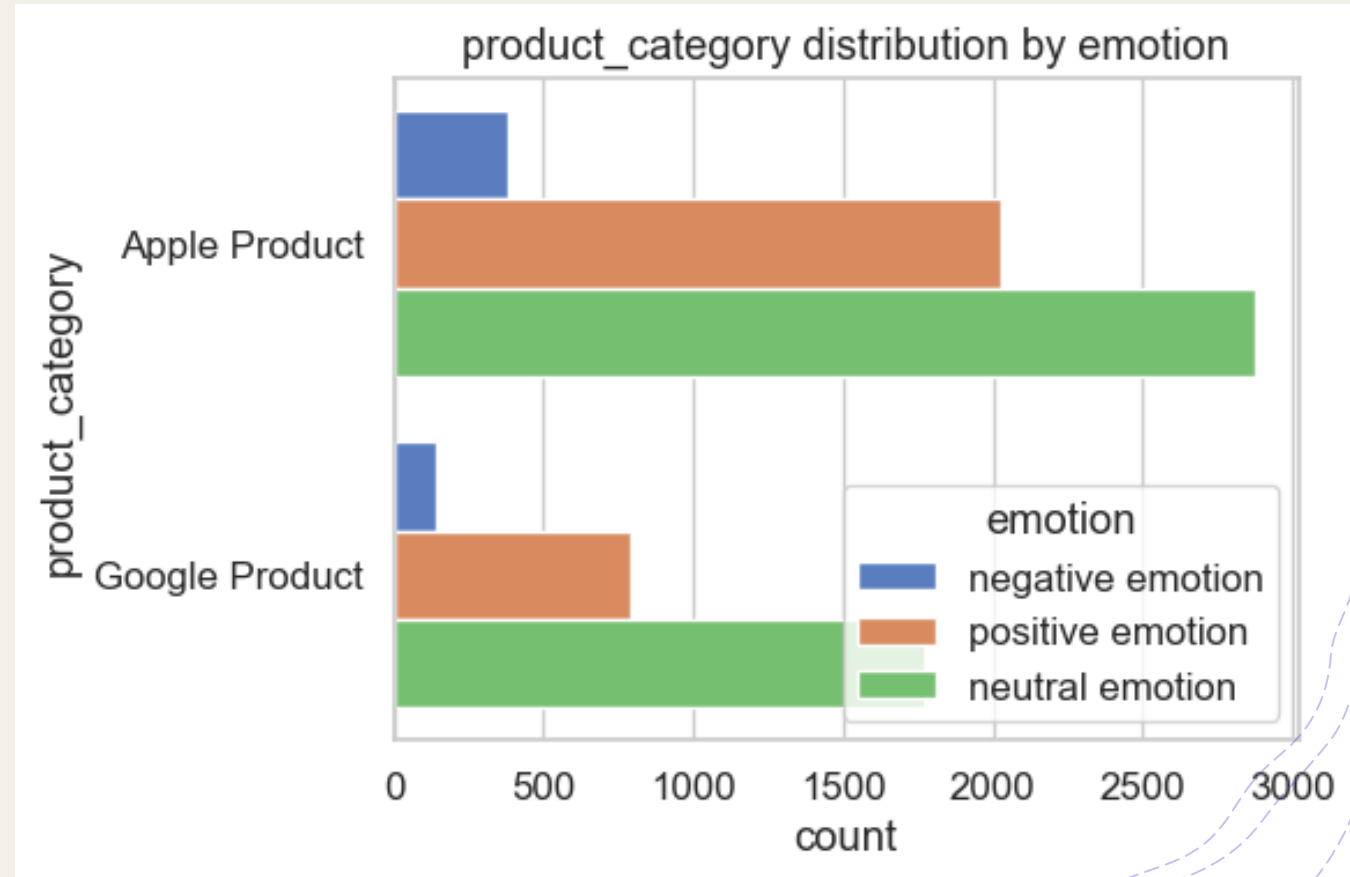
A terminal window showing the training progress of a model. At the top, a blue progress bar is at 100%, with the text "[252/252 51:32, Epoch 3/3]". Below this is a table with 6 columns: Epoch, Training Loss, Validation Loss, Precision, Recall, and F1. The table contains 3 rows of data for epochs 1, 2, and 3. At the bottom, another blue progress bar is at 100%, with the text "[84/84 02:43]". Below the progress bar is a line of text showing the final performance metrics and training parameters: "(0.978812484141081, {'lr': 3e-05, 'bs': 16, 'epochs': 4, 'max_len': 256})".

Epoch	Training Loss	Validation Loss	Precision	Recall	F1
1	0.621200	0.337938	0.755250	0.868350	0.789211
2	0.325800	0.278081	0.743055	0.895951	0.773311
3	0.172000	0.107899	0.919391	0.967901	0.941418

(0.978812484141081, {'lr': 3e-05, 'bs': 16, 'epochs': 4, 'max_len': 256})

Insights

- + Apple enjoys higher positive engagement
- + Google sees more neutral feedback; fewer emotional peaks
- + Consumer sentiment strongly driven by usability, design, and pricing
- + Negative emotion spikes align with major update issues or product releases



Conclusion

- + DistilBERT-based NLP model reached 90% accuracy
- + Automatically classifies public sentiment toward tech brands
- + Reveals key emotional and product themes driving consumer perception
- + Supports data-driven marketing and brand strategy

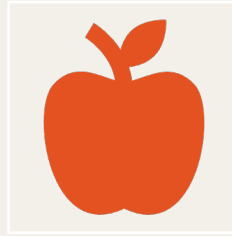


Recommendations



Operational:

Monitor sentiment trends post-launch via deployed dashboard



Marketing:

Apple: Emphasize innovation and premium experience
Google: Address user concerns about stability and privacy



Technical:

Retrain model quarterly with new tweets for evolving sentiment
Integrate topic modeling to extract emerging themes

Deployment

- + Platform: Streamlit App + Hugging Face Spaces
- + Features:
 1. Single text prediction
 2. CSV batch analysis
 3. Confidence visualization
- + Architecture:
 1. User → Streamlit UI → Tokenizer (tok) → DistilBERT Model → Output (Positive/Neutral/Negative)
- + Enables real-time public sentiment monitoring

[link to the sentiments App](#)



Tech Product Sentiment Analyzer

Analyze tweets about Apple & Google using a fine-tuned DistilBERT model.

Enter a tweet or short text:

Iphone is the best

Analyze Sentiment

Result:

Sentiment: Positive



Confidence: 86.16%

THE END

“Turning social media noise into business insight.”

Thank you!

